



THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
SCHOOL OF ELECTRICAL SYSTEMS ENGINEERING
DEPARTMENT OF INFORMATION & COMMUNICATION ENGINEERING
B. ENG. DEGREE EXAMINATIONS. FIRST SEMESTER 2024/2025 SESSION

COURSE CODE: ICT101 **COURSE TITLE:** Introduction to Information and Communication Technology
TIME ALLOWED: 2 Hours

UNITS: 1.0

Instruction: Read Carefully and Answer any Four (4) questions

Question 1

- Briefly explain the key historical milestones in the development of communication systems.
- Explain in brief, how has the evolution from 1G to 5G mobile communication systems impacted data speed, network capacity, and the types of services offered to users.

Question 2

- Explain briefly how technologies such as the telegraph, telephony, radio, satellite, data, optical, and mobile communications evolve over time.
- Discuss how the different types of transmission viz-a-viz analog vs. digital, wired vs. wireless affect the performance and reliability of communication systems?

Question 3

- Draw the block diagram of a typical communication system, and list the primary components involved in the transmission and reception of information.
- What are the advantages and disadvantages of guided transmission media like optical fiber, and how does it compare to unguided media in terms of performance and usage?

Question 4

- (i) Explain with reasons, when is unguided transmission media, such as radio waves or microwaves, be preferred over guided media, (ii) what are the key benefits of using unguided media?
- (i) How does multiplexing improve the efficiency of communication systems, (ii) state the differences between Frequency Division Multiplexing (FDM), Time Division Multiplexing (TDM), and Wavelength Division Multiplexing (WDM)?

Question 5

- Briefly explain the roles of Nigerian Communications Act and the Nigerian Communications Commission in regulating and overseeing the development of communication systems and technologies in Nigeria.
- (i) What are the main differences between analog and digital transmission in communication systems, (ii) explain the impact of these differences on signal quality and error handling.

Question 6

- Write a short note on how IT can help an organization in gaining a competitive advantage on its operation.
- Using a specific examples, describe how the information can be converted, stored, protected, processed, transmitted, and retrieved using computers, communication devices and software applications.